MIL-S-13985C
1 February 1983
SUPERSEDING
MIL-S-13985B
19 April 1974

MILITARY SPECIFICATION

SODIUM HYDROSULFITE, ANHYDROUS, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of anhydrous sodium hydrosulfite ($Na_2S_2O_4$).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	-	Strapping, Steel, and Seals
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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PPP-C-186 - Containers, Packaging and Packing For Drugs, Chemicals, and Pharmaceuticals
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-66 - Tape, Packaging, Vinyl Plastic Film

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

2.1.2 Other Government documents, drawings, and publicatons. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

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ASTM STANDARDS

C516 - Vermiculite Loose Fill Thermal Insulation
D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Physical characteristics. Anhydrous sodium hydrosulfite shall be a white or gray crystalline powder when tested as specified in 4.2.4.1.

3.2 Assay. Anhydrous sodium hydrosulfite shall contain no less than 90 percent by weight sodium hydrosulfite when tested as specified in 4.2.4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the anhydrous sodium hydrosulfite produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-2.

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4.2.2.2 For test. See 6.5 for sampling and testing precautions. Sampling for test shall be conducted in accordance with table I. A representative specimen of approximately 100 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE I. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using inspection level S-2 and an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Vermiculite cushioning missing or not as specified (level A only)
(f) Fiberboard pads and partitions missing or not as specified (where required)
(g) Marking incorrect, missing, or illegible
(h) Unitization not as specified

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. See 6.5 for sampling and testing precautions. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Physical characteristics. Visually examine approximately 50 g of the specimen for form and color.

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4.2.4.2 Assay.

(a) Formaldehyde solution. Add 50 milliliters (mL) of water to 25 mL of fresh, clear, 37-percent formaldehyde and adjust the pH to 8 to 9 by adding 0.3 g of sodium carbonate. Keep the solution in a dark bottle and do not permit its temperature to drop below 65°F.

(b) Procedure. Weigh to the nearest milligram approximately 8 g of the specimen into a small, dry weighing bottle. Pour 75 mL of the formaldehyde solution prepared in (a) into a 500-mL volumetric flask by means of a long-stemmed funnel. Remove the funnel carefully so as not to wet the neck of the flask. Put another dry long-stemmed funnel into the flask very carefully so that it does not touch the sides of the flask neck. Grasp the upper neck of the flask as well as the funnel and swirl the formaldehyde solution in a gentle circular motion. Pour the 8 g of specimen into the funnel in a steady stream but not fast enough to cause any appreciable accumulation of undissolved specimen. Continue swirling the flask until all the specimen has dissolved. Pour 10 mL of the formaldehyde solution prepared in (a) into the empty weighing bottle and then pour it into the flask, rinsing the bottle several times with water. Fill the flask to the volume mark with water and mix. Pipet 10 mL of this solution into a 250-mL beaker. Add 90 mL of water and 5 mL of 20-percent acetic acid. Titrate with 0.1N iodine solution adding 1-percent starch indicator solution near the end point. Calculate the percent by weight sodium hydrosulfite as follows:

$$\text{Percent sodium hydrosulfite} = 50\left(\frac{4.35AB}{W}\right)$$

where: A = Milliliters of iodine solution used in the titration,
B = Normality of the iodine solution, and
W = Weight of specimen in grams.

5. PACKAGING

5.1 Preservation. Sodium hydrosulfite shall be preserved level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. A quantity of 5.0 (+0.1 or -0) pounds (lb) of sodium hydrosulfite shall be unit packed in accordance with Department of Transportation (DOT) regulations in a glass bottle conforming to group A, class 1, type of glass optional, style 2, grade optional, closure A or R with seal optional of PPP-C-186. The bottle shall be cushioned on all exterior surfaces by a 1/2-inch minimum thickness of vermiculite, and placed upright within a metal can. The vermiculite shall conform to type I, size and density optional of ASTM C516. The can shall conform to type V, class 2 or type VI of PPP-C-96, coating plan B with side seam, if any, striped. The bottle and the can shall be closed

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tightly in accordance with the container manufacturer's instructions. The slip-cover can, if used, shall have the seam between the cover and body of the can sealed by nominal minimum 1-inch wide tape conforming to PPP-T-66.

5.1.1.2 Intermediate packing, level A. Four 5-lb containers of sodium hydrosulfite, unit packed as specified in 5.1.1.1, shall be intermediately packed upright in a close-fitting fiberboard box conforming to grade W5c of PPP-B-636. Full container height, close fitting cells shall be formed of half-slotted partitions for the containers. Additional pads shall be inserted in the pack as needed to prevent motion of contents. The box shall be closed as specified for the closure of interior containers in PPP-B-636.

5.1.2 Level C. A quantity of 5.0 (+0.1 or -0) lb of sodium hydrosulfite shall be unit packed level C in accordance with DOT regulations and in a manner to assure the maintenance of the specified purity of contents for a minimum period of six months and to assure acceptance by common carrier and safe delivery to destination.

5.2 Packing. Sodium hydrosulfite shall be packed level A, B or C as specified (see 6.2).

5.2.1 Level A. Two intermediate unit packs, prepared as specified in 5.1.1.2, shall be packed upright in a close-fitting wood box conforming to class 3, style optional, for a type 2 average load of PPP-B-585, or type overseas, style A, B, I or J of PPP-B-601. Motion of contents shall be prevented by insertion, where needed, of fiberboard pads formed from material conforming to grade V3c of PPP-F-320. The box shall be closed and reinforced by steel strapping as specified in the appendix to the applicable box specification. When flat steel strapping is used for Army shipments, it shall conform to finish A of QQ-S-781.

5.2.2 Level B. Two intermediate unit packs, prepared as specified in 5.1.1.2, shall be packed upright in a close-fitting fiberboard box conforming to grade V3c of PPP-B-636. Motion of contents shall be prevented by insertion, where needed, of fiberboard pads formed from material conforming to grade V3c of PPP-F-320. The box shall be closed and reinforced by pressure-sensitive adhesive filament tape banding or nonmetallic strapping as specified in the appendix to PPP-B-636.

5.2.3 Level C. A uniform quantity of containers of sodium hydrosulfite shall be packed upright in a box in a manner to assure safe delivery to destination. The box shall comply with DOT regulations, Uniform Freight Classification rules, and National Motor Freight Classification rules.

5.3 Unitization. For level A and level B protection, uniform quantities of packs of sodium hydrosulfite shall be palletized in accordance with the applicable requirements for load type I of MIL-STD-147 using steel strapping. The pallet shall conform to type IV of NN-P-71. For level A shipments, pallets

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and wood storage aids, including the wood cap, shall be treated with preservative as specified for the grade B pallet in NN-P-71. For level B shipments, a fiberboard cap shall be used. For level C shipments, palletization shall conform to DOT regulations and Uniform Freight Classification rules or National Motor Freight Classification rules, as applicable, and shall permit the stacking of palletized shipping containers in tiers at least 10 feet high without damage.

5.4 Marking. Marking shall be in accordance with DOT regulations and MIL-STD-129, and shall include lot, batch, or control number, as applicable, the manufacturer's part number, and the following precautionary information:

HAZARD
Flammable Solid
Irritant, Eye and Skin

WARNING! FLAMMABLE
Keep away from heat, sparks, open flame, acids, and oxidizers.
May ignite if allowed to become damp.
Store in a cool, dry place.
In case of fire, smother with soda ash, dry sand, or carbon dioxide.
Dispose of by flushing away with a large volume of water under adequate ventilation.
WARNING! CAUSES IRRITATION
Avoid contact with eyes, skin, and clothing.
Rubber gloves, chemical goggles, and protective work clothes should be worn to prevent contact during handling.
Wash thoroughly after handling.
FIRST AID: In case of contact, immediately flush eyes with plenty of water for at least 15 minutes.
Call a physician. Flush skin with water. (Wash clothing before reuse.)

6. NOTES

6.1 Intended use. Anhydrous sodium hydrosulfite is intended for use as a bleaching agent for laundries.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification, and the level of unit packing and packing required (see 5.1 and 5.2).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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6.5 Sampling and testing precautions. This specification covers inspection and use of chemical material which is potentially dangerous to personnel. Sodium hydrosulfite is a flammable solid and a dangerous fire hazard. Sodium bisulfite is an irritant of the eyes, skin, and mucous membranes. Such exposure should be avoided through use of appropriate personal protective equipment to prevent chemical contact with skin, eyes, and clothing. Formaldehyde, used in 4.2.4.2, is a known carcinogen. All applicable safety and occupational health rules, regulations, and procedures must be followed in the sampling and testing of this material.

Custodians:

Army - EA
Air Force - 68

Review activities:

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Preparing activity:

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5. PROBLEM AREAS

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